Mobile Crane Inspection Checklist (29 CFR 1910.180)

For frequent, periodic, idle-crane and standby-crane inspections, monthly rope checks and lift set-up.

OSHA 29 CFR 1910.180

DATE	SHIFT
INSPECTOR	AREA / EQUIPMENT ID

Before you start

1. Mark every item OK, Defect or N/A.
2. For any defect, have it carefully examined and decide whether it is a safety hazard, as 1910.180(d)(3) and (d)(4) require; record what was found and done.
3. The manufacturer's instructions and your site rules may add checks to this list.

Legal basis: 29 CFR 1910.180: Crawler locomotive and truck cranes (OSHA, Part 1910 Subpart N: Materials Handling and Storage) Source: eCFR, Title 29, text as of 2026-11-09

Online version and sources: https://safetyindex.blog/us/standards/1910-180-crawler-locomotive-truck-cranes

SECTION A | Operator, Records and Readiness

No.	Check item	What to look for	OK	Defect	N/A	Notes
1	**Designated operator** Ref. 1910.180(b)(3)	The person at the controls has been designated by the employer to operate this crane.	☐	☐	☐	
2	**Load rating chart** Ref. 1910.180(c)(2)	Durable chart with legible letters and figures fixed in the cab, readable from the seated control position.	☐	☐	☐	
3	**Monthly certification record** Ref. 1910.180(d)(6)	Record for brakes, hooks and ropes made this month, signed and dated, with the crane's serial number or other ID, and readily at hand.	☐	☐	☐	
4	**Monthly rope inspection record** Ref. 1910.180(g)(1)	Thorough rope inspection done within the past month; record on file with date, inspector signature and rope identifier.	☐	☐	☐	
5	**Idle crane (1 to under 6 months)** Ref. 1910.180(d)(5)(i)	A crane idle one month or more but under 6 months was inspected under (d)(3) and (g)(2)(ii) before returning to service.	☐	☐	☐	
6	**Idle crane (six months)** Ref. 1910.180(d)(5)(ii)	A crane idle for six months had a complete inspection covering (d)(3), (d)(4) and (g)(2)(ii) before returning to service.	☐	☐	☐	
7	**Standby crane** Ref. 1910.180(d)(5)(iii)	Standby crane given the (d)(3) and (g)(2)(ii) inspection at least semiannually.	☐	☐	☐	
8	**Idle rope approval** Ref. 1910.180(g)(2)(ii)	Rope that sat idle for a month or more was thoroughly inspected and approved before use, with a signed, dated record.	☐	☐	☐	
9	**Ballast or counterweight** Ref. 1910.180(i)(2)	Full ballast or counterweight specified by the maker in place and not exceeded; a truck crane that has dropped it is used only temporarily, with special care, for light loads.	☐	☐	☐	
10	**Fire extinguisher** Ref. 1910.180(i)(5)(i)	Carbon dioxide, dry chemical or equivalent extinguisher in the cab or near the crane.	☐	☐	☐	

General guidance only, not legal advice. Check the current regulation and your employer's procedures.

SECTION B | Frequent Inspection Items (Also Part of the Periodic Inspection)

No.	Check item	What to look for	OK	Defect	N/A	Notes
11	Control adjustment Ref. 1910.180(d)(3)(i)	Controls operate properly with no maladjustment; checked daily.	☐	☐	☐	
12	Control wear and contamination Ref. 1910.180(d)(3)(ii)	No excessive component wear; no lubricant or foreign matter fouling control mechanisms.	☐	☐	☐	
13	Safety devices Ref. 1910.180(d)(3)(iii)	Every safety device functions with no malfunction.	☐	☐	☐	
14	Air and hydraulic systems Ref. 1910.180(d)(3)(iv)	No leakage or deterioration in the air or hydraulic systems; checked daily.	☐	☐	☐	
15	Crane hooks Ref. 1910.180(d)(3)(v)	No cracks or deformation; throat opening no more than 15 percent over normal; twist no more than 10° from the plane of the unbent hook.	☐	☐	☐	
16	Rope reeving Ref. 1910.180(d)(3)(vi)	Reeving matches the manufacturer's recommendations.	☐	☐	☐	
17	Electrical apparatus Ref. 1910.180(d)(3)(vii)	Works properly; no signs of excessive deterioration and no dirt or moisture build-up.	☐	☐	☐	

SECTION C | Periodic Inspection Items

No.	Check item	What to look for	OK	Defect	N/A	Notes
18	Structure and boom Ref. 1910.180(d)(4)(i)	No deformed, cracked or corroded members anywhere in the structure or boom.	☐	☐	☐	
19	Bolts and rivets Ref. 1910.180(d)(4)(ii)	No loose bolts or rivets.	☐	☐	☐	
20	Sheaves and drums Ref. 1910.180(d)(4)(iii)	No cracked or worn sheaves or drums.	☐	☐	☐	
21	Pins, bearings, shafts and gears Ref. 1910.180(d)(4)(iv)	Pins, bearings, shafts, gears, rollers and locking devices free of wear, cracks and distortion.	☐	☐	☐	
22	Brake and clutch parts Ref. 1910.180(d)(4)(v)	No excessive wear on brake and clutch parts, linings, pawls or ratchets.	☐	☐	☐	
23	Indicators Ref. 1910.180(d)(4)(vi)	Load, boom angle and other indicators read accurately across their full range.	☐	☐	☐	
24	Power plant Ref. 1910.180(d)(4)(vii)	Gasoline, diesel, electric or other power plant performs properly and meets safety requirements.	☐	☐	☐	
25	Chain drives Ref. 1910.180(d)(4)(viii)	No excessive sprocket wear or chain stretch.	☐	☐	☐	
26	Travel steering, brakes and locks Ref. 1910.180(d)(4)(ix)	Travel steering, braking and locking devices work without malfunction.	☐	☐	☐	
27	Tires Ref. 1910.180(d)(4)(x)	No excessively worn or damaged tires.	☐	☐	☐	

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SECTION D | Wire Rope

No.	Check item	What to look for	OK	Defect	N/A	Notes
28	Rope diameter Ref. 1910.180(g)(1)(i)	No reduction below nominal diameter from loss of core support, corrosion or outside-wire wear.	☐	☐	☐	
29	Broken outside wires Ref. 1910.180(g)(1)(ii)	Any broken outside wires counted and their concentration judged not to be a hazard.	☐	☐	☐	
30	Outside wire wear Ref. 1910.180(g)(1)(iii)	No worn outside wires.	☐	☐	☐	
31	Wires at end connections Ref. 1910.180(g)(1)(iv)	No corroded or broken wires at end connections.	☐	☐	☐	
32	End connections Ref. 1910.180(g)(1)(v)	End connections not corroded, cracked, bent or worn, and correctly applied.	☐	☐	☐	
33	Kinks and damage Ref. 1910.180(g)(1)(vi)	No severe kinking, crushing, cutting or unstranding.	☐	☐	☐	
34	Limited-travel sections Ref. 1910.180(g)(2)(i)	Rope at equalizer sheaves, limited-travel sheaves and saddles checked closely for heavy wear and broken wires.	☐	☐	☐	
35	Nonrotating rope Ref. 1910.180(g)(2)(iii)	Nonrotating rope given a close, careful inspection.	☐	☐	☐	
36	Hoist rope Ref. 1910.180(h)(3)(ii)(a)	Hoist rope free of kinks before hoisting starts.	☐	☐	☐	
37	Multiple part lines Ref. 1910.180(h)(3)(ii)(b)	Part lines not twisted around each other.	☐	☐	☐	

SECTION E | Set-Up and Lifting

No.	Check item	What to look for	OK	Defect	N/A	Notes
38	Load within rating Ref. 1910.180(h)(1)(i)	Load, including blocks, hooks and slings, does not exceed the chart rating for the radius.	☐	☐	☐	
39	Crane level Ref. 1910.180(h)(3)(i)(a)	Crane is level and blocked where necessary.	☐	☐	☐	
40	Load secured and balanced Ref. 1910.180(h)(3)(i)(b)	Load well secured and balanced in the sling or lifting device before it goes higher than a few inches.	☐	☐	☐	
41	Outriggers Ref. 1910.180(h)(3)(ix)	Outriggers set whenever the load at the planned radius exceeds the manufacturer's no-outrigger rating.	☐	☐	☐	
42	Floats Ref. 1910.180(h)(3)(ix)	Any floats in use are securely attached to the outriggers.	☐	☐	☐	
43	Outrigger blocking strength Ref. 1910.180(h)(3)(ix)(a)	Wood blocks strong enough to prevent crushing.	☐	☐	☐	
44	Outrigger blocking condition Ref. 1910.180(h)(3)(ix)(b)	Wood blocks free from defects.	☐	☐	☐	
45	Outrigger blocking size Ref. 1910.180(h)(3)(ix)(c)	Wood blocks wide and long enough that they cannot shift or topple under load.	☐	☐	☐	
46	Hook position Ref. 1910.180(h)(3)(ii)(c)	Hook brought over the load so the load will not swing.	☐	☐	☐	

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SECTION E | Set-Up and Lifting

No.	Check item	What to look for	OK	Defect	N/A	Notes
47	No riders Ref. 1910.180(h)(3)(v)	No one rides the hook or the load during any hoisting, lowering, swinging or travel.	☐	☐	☐	

DEFECTS FOUND AND ACTION TAKEN

Item no.	Defect description	Reported to	Action taken / date closed

SIGN-OFF

SAFE TO OPERATE?	☐ Yes ☐ No: tagged out of service	
INSPECTOR SIGNATURE		DATE / TIME
SUPERVISOR REVIEW		DATE / TIME

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